Program

Day 1 (Aug 31)

Morning	
09:00	**Modeling Solar Wind Properties from the Lower Solar Atmosphere toward the SOLAR-C Era** Haruhisa Iijima
09:30	**Bridging the Corona and Heliosphere with Contemporary Solar Missions** Yeimy Rivera
09:50	**Spectroscopic diagnostics of coronal fan loops using AIA and EIS** Ananya Rawat
10:05	**A Solar Wind Connectivity Pipeline with Quantified Uncertainties: Application to Solar Orbiter's First High-Latitude Observations** Steph Yardley
10:20-11:05	**Break (20 min) / Poster (25 min)**
11:05	**Probing the coronal dynamics of the solar north pole using Proba-3, Hinode and Solar Orbiter.** Nawin Ngampoopun
11:20	**Analysis of full-disk EIS spectra over a solar cycle to determine the optimum EUVST/SoSpIM calibration method** Chloé Pilloud
11:35	**Origin of the solar magnetic field responsible for interplanetary magnetic field variations** Minami Yoshida
11:50-13:20	**Lunch (1 hour) /Poster (30 min)**
Afternoon	
13:20	**Heating of the chromosphere and corona in the active regions of the Sun** Souvik Bose
13:50	**Linking Chromospheric Wave Activity to Coronal Abundance Fractionation in a Solar Active Region Using Hinode/EIS and GST/FISS Observations** Kyoung-Sun Lee
14:10	**Multi-fluid and multi-species numerical simulations and their application to the FIP effect: Ebysus** Juan Martinez-Sykora
14:30	**Solar Spicules: Insights from 3D Radiative-MHD and Synthetic Diagnostics** Sanghita Chandra
14:45-15:05	**Break**
15:05	**Detection of UV Bursts in Solar Active Region Using Variational AutoEncoder** Masahiro Nishioka
15:20	**The connection between coronal abundances and chromospheric properties** Paola Testa

15:35	**Chromospheric dynamics and the O I 135.6 nm spectral line.** Viggo Hansteen
15:50	**Chromospheric Fe I lines in the near-ultraviolet solar spectrum** Edvarda Harnes
16:05	**Chromospheric Dynamics of an Umbral Flare Kernel from SCIP–DST Coordinated Observations** Ayumi Asai
16:20-17:30	**Poster**

Day 2 (Sep 1)

Morning	
09:00	**Solar Surface Magnetism: 20 Years Since Hinode's Breakthrough** David Orozco Suárez
09:30	**Two Decades of High-Resolution Spectropolarimetry of Solar Polar Magnetic Fields with Hinode/SOT-SP** Masahito Kubo
09:50	**Coronal Magnetic Field Modeling: From Hinode Achievements toward SOLAR-C** Daiki Yamasaki
10:10-11:00	**Break (20 min) / Poster (30 min)**
11:00	**Umbral Flashes and Magnetic Field Structures Revealed by Spectro-polarimetric Observations in He I 10830 Å** Yusuke Fukuchi
11:15	**Structure and evolution of solar prominences observed by Hinode/SOT during solar cycles 24 and 25** Stephane Regnier
11:30	**High-Resolution Sunspot Simulations and the Fine Structure of Penumbra and Evershed Flows** Hideyuki Hotta
11:45-13:15	**Lunch (1 hour) / Poster (30 min)**
Afternoon	
13:15	**First results from Sunrise III** Smitha Narayanamurthy
13:45	**Magnetic field distribution along spicules revealed with SUNRISE III/SCIP** Yoshihiro Naito
14:00	**Three-dimensional Magnetic Structures of Ellerman Bombs revealed by SUNRISE III/SCIP** Yusuke Kawabata
14:15	**DKIST observations of propagating chromospheric bright fronts** Catherine Fischer
14:30	**Multi-thermal Dynamics in Small-Scale Active Region Phenomena: Synergies from a Coordinated 2023 SOLARNET Ground-Space Campaign** Salvo Guglielmino
14:45-15:05	**Break**
15:05	**The SOLAR-C Mission in development** Toshifumi Shimizu
15:25	**Concept of Operations for the SOLAR-C Mission** Shin Toriumi
15:40	**Pre-flare and active region plasma flows and structure seen by the short wavelength camera on SOLAR-C/EUVST** James McKevitt

15:55	**Magnetic Reconnection Science Strategy for SOLAR-C** Shinsuke Imada
16:10-17:30	**Poster**

Day 3 (Sep 2)

Morning	
09:00	**Beyond Hinode: Coronal Heating Research with EUVST** Ignacio Ugarte
09:30	**Connecting Coronal Loop Heating Frequency to Magnetic Properties in Non-flaring Active Regions with Hinode/EIS** Will Barnes
09:50	**Effects of Plasma Composition Variations on the Radiative Cooling of Solar Flare Loops** Teodora Mihailescu
10:10-11:00	**Break (20 min) / Poster (30 min)**
11:00	**Power-law Relationship between Solar X-Ray Luminosity and Total Unsigned Magnetic Flux: From Active Regions through X-ray Bright Points** Keiji Yoshimura
11:15	**Multi-thermal analysis of loop morphology and restructuring in active regions** Bhinva Ram
11:30	**Coronal Rain Dynamics and Morphology under Different Coronal Heating and Magnetic Field Conditions** Takero Yoshihisa
11:45	**Differences in Heating Distribution between Active-region Warm Loops and Quiet-region CBP Loops** Hirohisa Hara
12:00	**Group photo**
12:30-18:00	**Excursion**
18:00-20:00	**Banquet**

Day 4 (Sep 3)

Morning	
09:00	**Modeling Elemental Abundance Variations in the Solar Corona** Jeffrey Reep
09:20	**Solar Soft X-ray Investigations Using CubeSat Missions** Crisel Suárez-Bustamante
09:40	**Towards a better understanding of coronal abundance anomalies: high-order multi-fluid modelling of active region loops** Nicolas Poirier
10:00-10:50	**Break (20min) / Poster (30 min)**
10:50	**Thermal and Magnetic Diagnostics of Small-scale Energy Release in Transient Brightenings of a Non-flaring Active Region Corona** Yuto Kondo
11:05	**Active-region upflows as a source of the slow solar wind: a statistical study using Hinode/EIS and IPS observations** Kazumasa Iwai
11:20	**The challenges of Solar Orbiter: what we have learned from an interplanetary solar physics mission so far** David Williams
11:35	**Spectropolarimetric Inversion in Four-Dimensions (SPIn4D): Progress and Prospect** Xudong Sun
11:50-13:15	**Lunch (1 hour) / Poster (25 min)**
Afternoon	
13:15	**Coronal heating by kink waves: RTV scaling laws & global models** Tom Van Doorsselaere
13:30	**RMHD Modeling of Reconnection Nanojets: Energy Budget, Spectral Signatures and Implications for the Solar-C Mission** Zekun Lu
13:45	**Spectral Line Ratios: A Diagnostic of Elemental Abundances** Luke Benavitz
14:00	**Faint Hot Components of Quiescent Active Regions: a Statistical Study with NuSTAR, XRT, and AIA** Jessie Duncan
14:15-15:05	**Break (20 min) / Poster (30 min)**
15:05	**Compression, Impact and Hot Rebound Flows from Coronal Rain Downflows** Patrick Antolin
15:20	**A Bayesian Approach to Assessing the Accuracy of Coronal Abundance Diagnostics** Natalia Zambrana Prado
15:35	**Hinode/EUV Imaging Spectrometer Observations of the Sun-as-a-Star** David Brooks

15:50-17:30	**Poster**

Day 5 (Sep 4)

Morning	
09:00	**Deciphering filament eruptions and flare energy release through coordinated observations with Solar Orbiter** Miho Janvier
09:30	**Fine structure of flares as revealed by Solar Orbiter/EUI** Hannah Collier
09:50	**Flare Fine Structure: Recent 3D Modeling Results and Opportunities for Solar-C** Joel Dahlin
10:10	**Spatial and temporal variations of energy transport during solar flares: looking forward to Solar-C** Graham Kerr
10:30-11:20	**Break (20 min) / Poster (30 min)**
11:20	**Solar flare footpoint line broadening is predominantly field-aligned** Andy Shu Ho To (Alexander Russell)
11:35	**Analyzing the hottest plasma in solar flares using X-ray measurements** Muriel Stiefel
11:50	**A long-term statistical analysis of cool coronal material linked to magnetic complexity and flaring** Luke McMullan
12:05-13:35	**Lunch (1 hour) / Poster (30 min)**
Afternoon	
13:35	**Modeling Flare Continuum Emission Observed by Hinode/EIS: Instrument Calibration and Element Composition Results** Peter Young
13:50	**Investigating Quasi-Periodic Pulsations in Solar Flares with Solar Orbiter/EUI Short-Exposure Images** Marie Dominique
14:05	**Comparing Observed and Synthetic Spectropolarimetric Signatures of Solar Flares with Machine Learning** Fabiana Ferrente
14:20	**Direct Focusing X-ray Imaging Spectroscopy of Solar Flares with FOXSI-4 and FOXSI-5** Shunsaku Nagasawa
14:35	**Characterizing Flare Eruptivity of Solar Active Regions Based on Photospheric Magnetic Properties and Coronal Modeling** Johan Muhamad
14:50	**Solar Flare Ion Temperatures** Alexander Russell
15:05-15:20	**Hinode Summary** Takashi Sakurai

Modeling Solar Wind Properties from the Lower Solar Atmosphere toward the SOLAR-C Era

*Haruhisa Iijima[1]

1. Nagoya Univ.

Understanding the origin of solar wind properties is a central problem in solar and heliospheric physics, linking plasma heating and acceleration in the solar atmosphere to the structure and variability of the heliosphere. In this talk, I will first provide a brief overview of solar wind research and its connection to the solar atmosphere. I will then present recent radiative MHD modeling efforts aimed at describing the solar wind from the lower solar atmosphere, focusing on how magnetic activity in and below the photosphere can influence coronal structure, open magnetic flux, and solar wind properties. Rather than treating the solar wind as a phenomenon determined only in the corona, these models provide a framework for examining how small-scale magnetic fields, chromospheric dynamics, and coronal-base processes are coupled across atmospheric layers. I will discuss selected results from recent and ongoing studies, including implications for the formation and variability of open-field regions and plasma outflows. Finally, I will discuss how SOLAR-C observations will help test these models and improve our understanding of the connection between the lower solar atmosphere and the solar wind.

Bridging the Corona and Heliosphere with Contemporary Solar Missions

*Yeimy Rivera[1]

1. Center for Astrophysics | Harvard & Smithsonian

Tracing the solar wind back to its solar sources has long been a central goal of the heliophysics community, pursued through decades of coordinated remote sensing and in situ observations. Parker Solar Probe has offered a unique vantage point for this purpose as it samples the solar wind in situ closer to its solar source than any previous spacecraft. This talk will highlight how this new era of conjunctions, multi-viewpoint imaging, and coordinated observing campaigns are used to investigate solar wind magnetic connectivity, source regions, and the evolution of structures as they propagate through the inner heliosphere. Specifically, I plan to discuss results from a comprehensive study tracing a single solar wind stream around the 2024 Total Solar Eclipse, which provided a unique opportunity for a synchronized, multi-perspective campaign across ground- and space-based observatories. The study integrated remote and in situ measurements to characterize the plasma and magnetic environment of an equatorial coronal hole from its coronal source to beyond the orbit of Mercury. I highlight the role of Hinode and other remote sensing observations in providing diagnostics of the solar atmosphere that complement inner heliosphere measurements from Parker Solar Probe and Solar Orbiter, illustrating how these missions together can build a more complete Sun-to-heliosphere picture. Additionally, I plan to discuss current limitations in multi-spacecraft coordination, offering insights to guide collaborative observational strategies for future heliophysics missions, such as Solar-C.

Spectroscopic diagnostics of coronal fan loops using AIA and EIS

*Ananya Rawat[1], Girjesh Gupta[1]

1. Udaipur Solar Observatory/ Physical Research Laboratory

We utilize spectroscopic and imaging data from the Extreme-ultraviolet Imaging Spectrometer (EIS) onboard Hinode and the Atmospheric Imaging Assembly (AIA) onboard the Solar Dynamics Observatory (SDO) to determine plasma and wave parameters along coronal fan loops. We determined the temperature along the loop lengths using the emission measure-loci (EM-loci) method using EIS spectral lines and found multi-thermal plasma components with temperatures ranging from 0.8-1.05, 1.2-1.4, and 1.6-1.8 MK. We further compared these temperature distributions along the loops with those derived from DEM analysis carried out using AIA coronal passbands. The multi-thermal nature of coronal fan loops can be well resolved using EIS spectroscopic data. On comparing the AIA 171 Å(0.8 MK) image with the EIS Mg V 276.16 Å(0.32 MK) image, we noted that low temperature coronal passbands (e.g., AIA 171 and 131 Å) have transition region contributions near loop footpoints. We determined the density along the loop using the EIS line-ratio method and AIA DEM analysis and found that it falls very slowly compared to the density fall expected from hydrostatic equilibrium. Our results demonstrate the multi-thermal nature of overdense loops, which can be well resolved with EIS spectroscopic data. We also determined the velocity amplitude of Alfvén waves from the non-thermal broadening of EIS spectral line profiles. We further estimated the Alfvén wave energy flux in the corona by combining these parameters with the newly reported fan loop magnetic field strength at the coronal footpoint ~200 G in Rawat et al. (2025). The newly estimated Alfvén wave energy flux is ~2×10^{7} erg cm^{-2} s^{-1}, significantly higher than those reported in earlier studies. The obtained Alfvén wave energy flux will contribute to coronal losses due to radiation along the fan loops in the active region.

A Solar Wind Connectivity Pipeline with Quantified Uncertainties: Application to Solar Orbiter's First High-Latitude Observations

*Steph Yardley[1,6], Adam Finley[2], Victoria Thompson[1], Nathanial Edward-Inatimi[3], Tristan Verge[4], Vasanth Anandan[5], Robert Wicks[1], Deb Baker[6]

1. Northumbria University, 2. ESA/ESTEC, 3. University of Reading, 4. Institut de recherche en astrophysique et plané tologie, CNRS, 5. Royal Observatory Belgium, 6. University College London Mullard Space Science Laboratory

Connecting in situ solar wind measurements with remote sensing observations of their coronal sources is critical for understanding solar wind origins and formation mechanisms. We present a connectivity pipeline that links in situ measurements to their source regions while accounting for solar wind acceleration. The pipeline combines HUXt (Heliospheric Upwind Extrapolation with time-dependence) and PFSS (potential field source surface) modelling, providing quantified uncertainties in the source location by accounting for variable source surface heights, multiple magnetogram inputs and perturbations in the spacecraft position. We apply the pipeline to the Solar Orbiter and Hinode/EIS observations taken during the first high-latitude perihelion in March 2025, during which the spacecraft performed a latitudinal scan across the southern polar coronal hole extension. From this, we determine probable source regions and characterise the properties of features within these well-connected regions (e.g. coronal bright points, jets) using EUI, SPICE, and Hinode/EIS observations. These are then linked to sub-stream structures in the in situ measurements using switchback detection methods and wavelet analysis. In the future, the connectivity pipeline will be made available to the community with applications for observation planning and space weather forecasting.

Probing the coronal dynamics of the solar north pole using Proba-3, Hinode and Solar Orbiter.

*Nawin Ngampoopun[1], Hardi Peter[1,2], Antoine Dolliou[1], Lakshmi Pradeep Chitta[1], Regina Aznar Cuadrado[1], Andrei Zhukov[3,4], Proba-3/ASPIICS Team

1. Max Planck Institute for Solar System Research, 2. Institut für Sonnenphysik (KIS), 3. Solar–Terrestrial Centre of Excellence —SIDC, Royal Observatory of Belgium, 4. Skobeltsyn Institute of Nuclear Physics, Moscow State University

The connection between coronal dynamics in the low and middle corona is not well understood, largely due to an observational gap caused by limitations of extreme ultraviolet telescopes and white-light coronagraphs. In this work, we report on continuous, coordinated, multi-spacecraft observations of the solar north pole made by Proba-3, Hinode, and Solar Orbiter (SolO) from 9 to 13 October 2025. During this observation period, a coronal hole and a large helmet streamer were observed at the north pole. Owing to the unique field of view of the ASPIICS coronagraph onboard Proba-3, we observe the coronal structure and dynamics at altitudes of 1.1–3 solar radii with high spatiotemporal resolution. The out-of-ecliptic viewpoint (~16° N) of Solar Orbiter also allows for better observation of the polar regions compared to Earth-based observatories. We identify various dynamic phenomena throughout the polar region, including density variations in the off-limb corona observed by Proba-3/ASPIICS, and coronal upflows near the coronal hole boundary observed by Hinode/EIS. In particular, we focus on whether these upflows are physically connected to the dynamics at the helmet streamer boundary, as the two regions are simultaneously captured within our combined fields of view. EUI/HRI and SPICE instruments onboard SolO further provide complementary information on the fine-scale dynamics and thermal structure of the outflow regions. Using these datasets, we aim to investigate the physical connection between coronal dynamics that are simultaneously observed in the low and middle corona and their possible relation to the dynamics of open-closed field boundaries and the origin of the solar wind.

Analysis of full-disk EIS spectra over a solar cycle to determine the optimum EUVST/SoSpIM calibration method

*Chloé Michelle Arlette Pilloud[1], Louise Harra[2], Toshifumi Shimizu[3], David Brooks[5], Krzysztof Barczynski[2], Yingjie Zhu[2], Shin Toriumi[3], James McKevitt[4], Marie Dominique[6]

1. ETH Zürich, 2. PMOD/WRC ETH Zürich, 3. JAXA/ISAS, 4. UCL, 5. NAOJ, 6. Royal Observatory Belgium

Full-Sun EUV spectroscopic observations are useful for understanding long-term coronal variability and will be a key component for supporting radiometric cross-calibration between SOLAR-C EUVST and SoSpIM. However, complete full-disk raster scans cost significant observing time, currently ˜40 hours for the EUV Imaging Spectrometer (EIS) on Hinode, limiting how frequently they can be obtained.

Here we use archival full-disk rasters from Hinode/EIS as reference observations and systematically reduce them to a range of sparse sampling datasets. We compare disk-integrated line intensities from these reduced rasters with the corresponding full-disk values (including SDO/AIA) to quantify the reconstruction accuracy, and determine what level of sparse spatial sampling is sufficient for future calibration campaigns. By applying this approach to observations taken throughout the solar cycle, we are also able to examine how sampling performance depends on solar activity, spatial distribution, and the presence of active regions or the activity belt.

Our results show that the EIS raster step size can be increased significantly while preserving the full-Sun intensity estimate over the solar cycle. For Fe XII 195.12A, 16 full-Sun datasets from 2013–2025 were tested with sparse factors n = 1–50, where n = 1 corresponds to the original 4 arcsec spacing in the raster direction. Sampling up to n = 10, or 40 arcsec, reproduces the full-resolution intensity to within 8%, with an average deviation of about 4.5%, while n = 5–9 remains within 1–3%. Comparable results are found for other spectral lines such as Fe X 184.54A and Fe XV 284.16A. The results imply that even quite large step sizes are sufficient to sample the coronal intensity distribution, and that a full EIS scan could potentially be shortened from ˜40 hours to ˜8 hours or less while retaining the accuracy needed for future LYRA, SoSpIM, and SOLAR-C/EUVST cross-calibration campaigns.

Origin of the solar magnetic field responsible for interplanetary magnetic field variations

*Minami Yoshida[1], Toshifumi Shimizu[1], Shin Toriumi[1], Haruhisa Iijima[2]

1. ISAS/JAXA, 2. Nagoya University

The global distribution of the solar magnetic field governs the structure of the solar wind and the interplanetary magnetic field (IMF). Understanding the connection between photospheric magnetic structures and the IMF is crucial for identifying solar wind source regions. While the contribution of individual active regions to open flux is theoretically discussed, observational verification over the solar cycle remains a challenge. We addressed this through two approaches using SDO/HMI data. First, we simulated the magnetic environment of 2014 using a surface flux transport (SFT) model. The results indicate that the rapid increase in IMF was driven by the merging of open fields from sunspot groups with pre-existing coronal holes, a process that enhances the equatorial dipole component through differential rotation. Second, numerical experiments across Cycle 24 revealed a characteristic behavior during polarity reversal. While open flux around sunspot groups is expected to increase when the polar field is weak, observational data show a stagnant trend. We attribute this to the emergence of multiple sunspot groups forming closed magnetic loops that suppress the extension of open flux. Our findings suggest that IMF variations during solar maximum are determined by the competition between two processes: the enhancement of open flux via the merging of background fields and its suppression by closed loops from complex sunspot configurations. These findings provide a key point for the SOLAR-C mission by identifying the changes in magnetic fields as an observational target. We suggest that the evolution of these magnetic structures at solar wind sources should be observed for high-resolution and high-sensitivity observations with EUVST to clarify the origin of solar wind variability.

Heating of the chromosphere and corona in the active regions of the Sun

*Souvik Bose[1]

1. University of Oslo

The solar chromosphere serves as the critical interface through which magnetoconvective energy must pass to heat the million-degree corona. While 3D radiative-MHD simulations and high-resolution observations have revealed a host of candidate heating mechanisms--ranging from waves and shocks to flux emergence and reconnection, identifying which processes dominate remains a major challenge. In this talk, I will present recent progress in sharpening this picture by confronting coordinated ground- and space-based observations with state-of-the-art numerical models. Our results demonstrate that heating in active-region plage is tightly coupled across atmospheric layers down to scales of a few hundred kilometers, driven by the dissipation of currents from magnetic braiding. Crucially, our recent work pinpoints the locus of this energy release: it occurs not within the strong-field cores of magnetic flux tubes, but along their weaker-field edges. This strongly implicates magnetic geometry, rather than field strength alone, as the governing factor in atmospheric heating. I will conclude by discussing how upcoming capabilities with DKIST, SOLAR-C, and MUSE will allow us to put these geometric heating models decisively to the test.

Linking Chromospheric Wave Activity to Coronal Abundance Fractionation in a Solar Active Region Using Hinode/EIS and GST/FISS Observations

*Kyoung-Sun Lee[1]

1. Seoul National University

Elemental abundance fractionation in the solar corona is a key diagnostic of how plasma is processed between the chromosphere and the corona. The First Ionization Potential effect is thought to be driven by ponderomotive forces associated with magnetohydrodynamic waves in the partially ionized chromosphere. However, direct observational evidence linking chromospheric wave activity to coronal abundance variations remains limited. In this talk, I will present recent results on the connection between chromospheric waves and coronal abundances in an active region. By combining Hinode/EIS measurements of coronal composition with high-resolution chromospheric spectroscopy from GST/FISS, and by using SDO/AIA and SDO/HMI data for coalignment and magnetic context, we examine whether spatial variations in chromospheric wave properties are associated with the observed structure of coronal abundances. These observations provide a possible observational link between lower atmospheric wave activity and the spatial structuring of coronal abundances. We found that chromospheric transverse waves are mainly detected near loop footpoints, particularly along the sunspot penumbra and superpenumbral fibrils. These wave locations are magnetically connected to enhanced coronal abundance fractionation, especially along closed magnetic fields. The detected waves are dominated by periods of several minutes, and inward (downward) propagating low-frequency waves appear to be a significant component. These properties are consistent with the idea that transverse MHD waves can contribute to FIP fractionation through ponderomotive forces in the chromosphere. I will briefly discuss remaining observational challenges, including the need for higher cadence and spectroscopic observations at multiple heights to better resolve wave propagation, reflection, and possible resonant behavior. Such observations are important for clarifying how chromospheric wave dynamics shape coronal plasma composition.

Multi-fluid and multi-species numerical simulations and their application to the FIP effect: Ebysus

*Juan Martinez-Sykora[1]

1. SETI institute

The chromosphere is partially ionized, transitions from a collisional to a weakly collisional regime, becomes magnetically dominant, and could be the site of chemical fractionation producing the so-called FIP effect –the fractionation of elemental abundance based on the first ionization potential. We have been developing a multi-fluid, multi-species numerical code called Ebysus, designed to address these challenges self-consistently. The code treats each excited/ionized level of each desired species as a separate fluid, self-consistently including physical processes such as thermal conduction, ion coupling, non-equilibrium ionization/recombination, collisions, and radiation. In this work, we describe recent results, capabilities, and perspectives on the use of Ebysus for the FIP effect and chemical fractionation.

Solar Spicules: Insights from 3D Radiative-MHD and Synthetic Diagnostics

*Sanghita Chandra[1], Robert H. Cameron[1], Damien F. Przybylski[1], Sami K. Solanki[1]

1. Max Planck Institute for Solar System Research

What are spicules? Interpreting their observational appearance, is challenging due to line-of-sight superposition, Doppler-dependent visibility effects, and the difficulty of reconstructing three-dimensional (3D) structures from two-dimensional (2D) data. Recent advances in realistic 3D radiative-MHD simulations provide a means to overcome these limitations.

Using an enhanced network simulation with MURaM-ChE and a novel Hα proxy that captures the full 3D opacity structure, we investigate the evolution of spicules and their on-disc counterparts—rapid blue- and red-shifted excursions (RBEs/RREs). A detailed case study of an RBE shows that it is driven by flux emergence and magnetic reconnection, which build pressure gradients that launch the jet. We also identify an associated, faster heating front propagating at Alfv´enic speeds.

We further perform a statistical analysis of 58 off-limb spicules (types I and II) and compare their properties with previous high-resolution observations in Ca II H and Hα. The synthetic spicules exhibit morphological characteristics and lifetimes broadly consistent with observations. Analysis of their 3D structure reveals predominantly thread-like and slab-like geometries, and shows that RBEs and spicules share a common underlying opacity structure. The very high apparent speeds (~190 km/s) inferred for some spicules arise from an optical effect rather
than representing actual mass upflows. We additionally find that spicules preferentially form near quasi-separatrix layers (QSLs), underscoring the importance of magnetic topology in driving their dynamics. To probe their thermal evolution, we analyse synthetic Si IV (transition region) and HRI/EUV (coronal) emission, investigating whether spicules persist in hotter channels after fading from Hα, leveraging the full 3D information available in the simulations.

Detection of UV Bursts in Solar Active Region Using Variational AutoEncoder

*Masahiro Nishioka[1,2], Shin Toriumi[2,1], Yusuke Iida[3]

1. The Graduate University for Advanced Studies, SOKENDAI, 2. Institute of Space and Astronautical Science/Japan Aerospace Exploration Agency, 3. Niigata University

The solar transition region is a thin region between the chromosphere and the corona, where temperatures rise sharply. Detailed diagnostics of the UV spectra originating from this region using Interface Region Imaging Spectrograph (IRIS) have revealed the existence of small, sudden brightening events (UV Bursts) in solar active regions. Previous studies have suggested that the magnetic reconnection mechanism that causes UV Bursts may not be uniform. Therefore, it is necessary to classify events commonly referred to as UV Bursts into different physical classes based on their spectral shapes and to statistically clarify the magnetic and plasma conditions associated with each. In this study, we focused on solar active region 11850 on September 24, 2013, where the presence of UV Bursts was reported by Peter et al. (2014), and attempted to detect UV Bursts from IRIS spectroscopic observation data using Variational AutoEncoder (VAE), a machine learning model. VAE is a neural network that represents input data as probability distributions in a latent space and reconstructs outputs of the same size through sampling. It is used for anomaly detection with large reconstruction errors. The training data for VAE consisted of independent samples of one-dimensional spectra for each spatial pixel at every time point. We created snapshots for each slit in the scan order, and detected anomalous spectra from the image differences between input and output. Among the pixels detected as anomalous spectra, almost all large areas showed significant brightness changes and broadened profile widths. This is consistent with the characteristics of UV Bursts described by Young et al. (2018). Furthermore, the detected anomalous spectra were classified according to their properties by correlating them with magnetic field data. In this presentation, we will discuss the physical characteristics of UV Bursts and the potential applications of this method in next-generation solar observations.

The connection between coronal abundances and chromospheric properties

*Paola Testa[1], Juan Martinez-Sykora[3,2,4,5], Bart De Pontieu[2,4,5], Alberto Sainz Dalda[3,2], Viggo Hansteen[3,4,5]

1. Harvard-Smithsonian Center for Astrophysics, 2. Lockheed Martin Solar and Astrophysics Laboratory, 3. SETI, 4. Rosseland Centre for Solar Physics, University of Oslo, 5. Institute of Theoretical Astrophysics, University of Oslo

Elemental abundances in the solar corona and solar wind are often observed to differ from those in the solar photosphere, most commonly showing an enhancement of low first ionization-potential (FIP) elements (the FIP effect). The observational evidence of the connection between the chemical fractionation in the solar atmosphere with FIP suggests that the mechanisms responsible for this effect take place in the chromosphere, where low-FIP elements are mostly ionized, while high-FIP elements remain mostly neutral. We discuss the findings of recent observational studies using Hinode, IRIS, and SDO that have investigated the footprint of coronal abundance anomalies in the lower atmosphere, which can provide much needed constraints on the fractionation mechanisms. We also discuss how EUVST can significantly advance these investigations, as well as the new techniques we use for the abundance derivation from spectral data.

Chromospheric dynamics and the O I 135.6 nm spectral line.

*Viggo Hansteen[1,2]

1. SETI Institute/LMSAL, 2. RoCS/University of Oslo

The O I 135.6˜nm spectral line is formed in the chromosphere at the same heights as the Mg II h&k line cores are formed. As the O I line is optically thin, it represents a possibility for measuring the non-thermal velocities in this region without the complications added by optically thick radiative transfer. Numerical models have hitherto strained to reproduce Mg II core line widths, challenging current understanding of chromospheric energetics and dynamics.
A set of numerical models of varying resolution, size, magnetic topology and strength are considered and used to synthesize O I line emission and to investigate the constraints that observations of this line place on chromospheric dynamics and densities. We find that, for quiet Sun, while non-thermal motions undeniably provide a source of Doppler broadening and chromospheric mass loading, the average strength of the photospheric magnetic field is the most important parameter in setting the Mg II core width to values within 5 km of observed values. Furthermore, for plage, we identify non-equilibrium hydrogen ionization and three dimensional radiative transfer as important ingredients in understanding chromospheric diagnostics and deciphering chromospheric structure.

Chromospheric Fe I lines in the near-ultraviolet solar spectrum

*Edvarda Harnes[1,2], Smitha Narayanamurthy[1], Damien Przybylski[1], Robert Cameron[1], Andreas Lagg[1], Sami Solanki[1]

1. Max Planck Institute for Solar System Research, 2. Georg-August-Universität Göttingen

In the near-ultraviolet (NUV) and blue part of the solar spectrum there are several Fe I lines that show very broad profiles typical of chromospheric lines. The diagnostic potential of these spectral lines is largely unexplored due to a lack of high-resolution observations. With the successful flight of the SUNRISE III balloon-borne observatory we have for the first time full spectro-polarimetric data at high spectral and spatial resolution in this wavelength region, overlapping with the spectral region accessible by ground-based observatories around 400 nm. We have investigated the formation properties of four of these lines observed by the SUSI instrument onboard SUNRISE III. The cores of these lines form in the chromosphere in the height range between the Ca II 854.2 nm line and the Mg I b2 517.2 nm line and can provide additional constraints on magnetohydrodynamic (MHD) simulations. We will present results from line synthesis in 3D radiation MHD simulations made with MURaM and the effect of simulation resolution on the averaged and spatially resolved spectral lines.

Chromospheric Dynamics of an Umbral Flare Kernel from SCIP–DST Coordinated Observations

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We present spectroscopic observations of an M1.4 flare obtained through a coordinated observing campaign between the SUNRISE-III/SCIP and Domeless Solar Telescope (DST) at Hida Observatory, Kyoto University. The flare occurred in the NOAA Active Region 13738 and exhibited a compact flare kernel within a sunspot umbra. Our data reveals the temporal and morphological evolution of the kernel in the umbra. Spectral diagnostics in H-alpha, Ca II 8542, and Na I D1/D2 lines taken with DST and Ca II 8542, K I D1/D2 lines taken with SCIP enable us to examine the features of flare-induced brightenings and to quantify plasma motions. High-resolution SCIP data resolves fine structures that are smeared in DST observations, offering new insights into the small-scale processes of flare energy release. In this presentation, we will present observational data from SCIP and DST, and will discuss the energy injected into the flare core and the plasma dynamics occurring there.

Solar Surface Magnetism: 20 Years Since Hinode’ s Breakthrough

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Since the launch of *Hinode* nearly two decades ago, the study of solar magnetism from space at high spatial resolution has profoundly transformed our understanding of the Sun. *Hinode* set a benchmark by delivering unprecedented observations that combined fine spatial detail with stable and highly precise spectropolarimetric measurements. Over the past 20 years, matching this unique combination, high spatial resolution together with consistently reliable polarimetric accuracy, has remained a major observational challenge. In recent years, new facilities have opened a transformative era for solar physics. Ground-based observatories such as the Daniel K. Inouye Solar Telescope (DKIST), together with space- and balloon-borne instruments like *SUNRISE III* and the Polarimetric and Helioseismic Imager (SO/PHI) on board *Solar Orbiter*, in particular its High Resolution Telescope (HRT), are now pushing the limits of spatial resolution, sensitivity, and diagnostic capability. These instruments provide complementary views that are reshaping our understanding of solar magnetism across different spatial and temporal scales. Among the broad range of scientific objectives enabled by these facilities, in this contribution I will focus on recent advances in the observation of the quiet Sun. In particular, the data obtained with the SCIP instrument on board *SUNRISE III* represent a substantial step forward. These observations reveal previously unresolved magnetic structures with an unprecedented combination of spatial resolution and polarimetric sensitivity, offering new insights into the fine-scale organization, dynamics, and energetics of quiet-Sun magnetism.

Title: Two Decades of High-Resolution Spectropolarimetry of Solar Polar Magnetic Fields with Hinode/SOT-SP

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High spatial resolution, high-precision spectropolarimetry with Hinode/SOT-SP has transformed our view of the Sun's polar regions. These measurements have enabled, for the first time, detailed studies of the photospheric vector magnetic field at high latitudes. Polar observations are important for understanding solar cycle variability, and polar magnetic flux is a key precursor of the next cycle's strength because it correlates with the subsequent maximum sunspot number. SOT-SP photospheric vector magnetic field measurements reveal the properties of kilogauss magnetic patches and ubiquitous horizontal fields in the polar regions. Hinode has observed the polar regions regularly under stable conditions. This enables us to construct polar panorama maps for nearly two solar cycles and to track the polarity reversal with unprecedented detail. These synoptic observations indicate that the polarity reversal is driven predominantly by magnetic patches with large magnetic flux density, such as kilogauss patches. Despite this progress, precise polar field measurements remain challenging, mainly because observations from the ecliptic plane suffer from severe foreshortening and unfavorable viewing geometry. We find that the magnetic flux derived from SOT-SP vector field measurements tends to be underestimated in the polar regions. Measurements from outside the ecliptic are now available through the Solar Orbiter mission. This provides a timely opportunity to reassess what we have learned from SOT-SP, to connect Hinode and Solar Orbiter diagnostics, and to identify remaining issues in observing and modeling polar fields. In this talk, we will summarize key Hinode/SOT-SP results on the polar magnetic field and discuss future prospects for polar field studies.

Coronal Magnetic Field Modeling: From Hinode Achievements toward SOLAR-C

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1. Japan Aerospace Exploration Agency

The Hinode mission has revolutionized our understanding of solar magnetic activity through its high-precision spectropolarimetric observations of the photospheric magnetic field. In particular, observations from the Solar Optical Telescope (SOT) have provided unprecedented insights into the magnetic energy build-up processes leading to solar flares and have enabled the identification of small-scale magnetic structures that trigger the onset of eruptive events. In parallel with observational advances, significant progress has been made in three-dimensional coronal magnetic field modeling. Nonlinear force-free field (NLFFF) extrapolation techniques, using photospheric magnetic field measurements as boundary conditions, have become a powerful tool for investigating the non-potential magnetic structures responsible for energy storage and release in the solar atmosphere. Looking ahead to the SOLAR-C era, high-cadence ultraviolet spectroscopic observations covering a broad temperature range from the chromosphere to the corona are expected to provide unprecedented diagnostics of plasma dynamics and magnetic activity throughout the solar atmosphere. However, direct measurements of coronal magnetic fields remain challenging, making magnetic field modeling an essential component for interpreting these observations and understanding the underlying magnetic structures. To address the limitations of the force-free assumption adopted in conventional NLFFF modeling, magnetohydrostatic (MHS) extrapolation techniques have recently been developed to incorporate the effects of plasma pressure and gravity. In this invited talk, I will review the achievements of magnetic field observations and coronal magnetic field modeling during the Hinode era, present recent progress in MHS magnetic field extrapolation, and discuss how the synergy between advanced magnetic field modeling and future SOLAR-C observations will open new opportunities for exploring the magnetic coupling of the chromosphere and corona.

Umbral Flashes and Magnetic Field Structures Revealed by Spectro-polarimetric Observations in He I 10830 Å

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Umbral Flashes (UFs) are transient chromospheric brightenings observed above sunspot umbrae, with a typical period of about three minutes, and are associated with slow-mode MHD waves. Through spectro-polarimetric observations of UFs in chromospheric lines, periodic variations in radiative intensity, line-of-sight (LOS) velocity, and magnetic field have been reported. However, the relationship between UFs and magnetic fields, as well as their connection to movements such as upflow/downflow, remains unclear. We conducted near-infrared spectro-polarimetric observations of a sunspot umbra in the NOAA Active Region 14064 on April 21, 2025 using Domeless Solar Telescope at Hida Observatory, with a fixed slit position and a time cadence of 8 s for approximately one hour. We determined the zero-crossing wavelength of the Stokes-V profiles near the He I 10830 Åline center to derive time series of the line-center intensity and line-of-sight (LOS) velocity. We found that blueshifts tended to synchronize with brightenings, with intensity enhancements slightly preceding the velocity signals, with periods of about 2-3 minutes. In addition, propagating patterns along the slit with similar periods were detected even in the Stokes-Q and U signals. Furthermore, the wavelength-integrated absolute Stokes-V signal around the He I 10830 Åline center tended to decrease during particularly strong brightenings. Based on these observational results, we discuss the physical interpretation of UFs and the magnetic field structure in the observed sunspot.

Structure and evolution of solar prominences observed by Hinode/SOT during solar cycles 24 and 25

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1. Northumbria University

Solar prominences are structures suspended in the corona confining relative cool plasma into a coherent magnetic structure. The properties and statistics of prominences are known to vary the solar cycle, especially the distribution in latitude, the orientation of the magnetic field, and their apparent fine structure. However, no comprehensive studies have been yet performed with the high-spatial resolution, high cadence, and long-term observations that can provide the Solar Optical Telescope (SOT) on board the Hinode mission. We select a large sample of prominences in the Halpha and Ca II H spectral lines covering solar cycle 24 (2008-2019) and most of solar cycle 25 (2019-present). We study the properties of the fine structure of prominences using the Rolling Hough Transform (RHT, see Birch and Regnier 2025, ApJ, 985, 161), and thus providing the statistics for: (i) single snapshots during the solar cycle, (ii) time series for quiet and active prominences. We clearly identify the change in structure during eruptions as well as the location of plasma draining. We discuss our results in terms of the global dynamo process (e.g. meridional flow and quiet-Sun evolution).

High-Resolution Sunspot Simulations and the Fine Structure of Penumbra and Evershed Flows

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High-resolution simulations have successfully reproduced part of the sunspot penumbra and the Evershed flow. Sunspots are known to consist of an umbra with low brightness and strong vertical magnetic fields, and a relatively bright penumbra with strong horizontal magnetic fields. Furthermore, the penumbra contains an outward Evershed flow. These penumbral structures and flows have been extensively studied by high-resolution solar observations, but their physical origin remains unclear. For example, Rempel (2012) reproduced the Evershed flow and penumbra by imposing a horizontal magnetic field, but the realism of this setting is unclear, and discrepancies with observations have been pointed out.
This study investigated the effects of increasing resolution to explore the origins of the penumbra and the Evershed flow with a potential magnetic field boundary condition. While sunspots were typically computed with a grid spacing of 30–40 km, we successfully simulated slab-type sunspots with a grid spacing reduced to 3 km and circular sunspots to 6 km. Using a 6 km grid, we successfully reproduced an Evershed flow with an average speed of about 2.5 km/s and locally exceeding 10 km/s. Furthermore, the high-resolution calculations clearly reproduced the penumbra.
We analyzed the simulation result and identified the physical reason why these structures can be reproduced in the high-resolution simulation. The primary physical mechanism accelerating the Evershed flow is almost identical to that in previous works: the Lorentz force redirects upflows to the Evershed flow. An essential difference between different resolutions is the upflow structure in penumbral regions. In the high-resolution simulation, small-scale elongated fast upflows exist that provide significant mass to the Evershed flow. These upflows can only be properly resolved in high-resolution simulations, and consequently, the Evershed flow appears only at higher resolution as well. The reproduced upflow structure is small, with a typical scale of several 100 km.

First results from Sunrise III

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In July 2024, Sunrise III flew from Sweden to Canada at stratospheric heights capturing high resolution diffraction-limited images of the Sun for 6.5 days. The telescope carried three scientific instruments each complementing the other by observing in different parts of the solar spectrum from the near-ultraviolet by SUSI (309 nm - 417 nm), the visible by TuMag (517 nm - 525 nm) to the near infrared by SCIP (765 nm - 855 nm). During its flight, Sunrise III observed a wide variety of features on the Sun including the quiet regions, sunspots, plages, filaments, spicules, flux emergence, and also flares. By combining the data from the three instruments, it is possible to infer and understand various properties of the solar atmosphere from the photosphere to the upper chromosphere. In this talk, I will present highlights from Sunrise III flight and discuss some of the first results from the mission.

Magnetic field distribution along spicules revealed with SUNRISE III/SCIP

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Understanding the magnetic field structure of spicules is essential for developing formation models of spicules, which are important for understanding the mass transport to the corona and the coronal heating. We report the spatial distribution of spicule magnetic fields from spectropolarimetric observations with SUNRISE-III/SCIP, which provides high spatial resolution and high polarization sensitivity under seeing-free conditions. We observed the solar limb above a quiet region with SCIP and detected clear Stokes V signals in the Ca II 8542 Å line from off-limb spicules. We applied the Weak Field Approximation to estimate the line-of-sight component of the magnetic field (B_{LOS}) of spicules. We find that B_{LOS} ranges up to 20 G, nearly independent of height, in the region up to 2 Mm from the limb, whereas the distribution broadens to 40 G at 2-4 Mm from the limb. The Stokes I profiles averaged over successive distance intervals from the limb reveal a transition from a double-peaked shape to a single-peaked one at around 2 Mm from the limb. This suggests that the observed B_{LOS} depends strongly on the optical thickness of the Ca II lines. In the optically thick region with double-peaked profiles, the estimated B_{LOS} represents a superposition of a few foreground spicules, whereas in the optically thin region with single-peaked profiles, the estimated B_{LOS} reflects contributions from a number of prominent, tall, inclined spicules, leading to more dispersed B_{LOS} values.

Three-dimensional Magnetic Structures of Ellerman Bombs revealed by SUNRISE III/SCIP

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Ellerman bombs (EBs) are transient brightenings in the wings of chromospheric lines, recognized as signatures of magnetic reconnection in the lower solar atmosphere. However, the three-dimensional (3D) magnetic topology of EBs has remained elusive due to the lack of seamless height coverage in spectropolarimetric observations. We present initial results from SUNRISE III/SCIP observations of an emerging flux region obtained on 2024 July 15. SCIP performed simultaneous full-Stokes spectropolarimetry in Fe I 846.8 nm, K I D1/D2 (769.8/766.4 nm), and Ca II infrared triplet lines (849.8/854.2 nm), providing seamless diagnostics from the photosphere to the chromosphere with diffraction-limited spatial resolution. By applying the Weak Field Approximation (WFA) to the multi-line dataset, we reconstructed the three-dimensional line-of-sight magnetic field. We identify two distinct classes of reconnection events, illustrated by examples Event 1 and 2. In Event 1, the bipolar magnetic structure is confined to the lower layers (K I and Ca II wing heights), while a unipolar field configuration and no significant core intensity enhancement are found in the Ca II 854.2 nm core. These results indicate the low-altitude reconnection and heating. In Event 2, the bipolar structure persists up to the Ca II 854.2 nm core formation height, accompanied by a clear core brightening, indicating that the current sheet extends to higher altitudes and drives chromospheric heating. These contrasting behaviors were predicted by MHD-based synthetic observations (Kawabata et al. 2024), and the SCIP observations now provide direct observational confirmation. These results demonstrate that SCIP has successfully resolved the 3D structure of magnetic reconnection at different atmospheric heights. Future work will employ non-LTE inversions to derive the thermal structure.

DKIST observations of propagating chromospheric bright fronts

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We present high-cadence chromospheric image sequences recorded with the Daniel K. Inouye Solar Telescope (DKIST) that reveal propagating arc-shaped bright fronts of a few hundred kilometers width. These bright fronts maintain their arc shape for up to one minute and originate from chromospheric bright grains seen at the center of the arc. We identify photospheric G-band bright points underneath the chromospheric bright grains, which are typically the signature of small-scale magnetic flux concentrations. The G-band bright points are found to interact with co-located vortices prior to the appearance of the chromospheric bright fronts. Images synthesized from three-dimensional magnetohydrodynamic simulations show a striking similarity with the discovered phenomenon and the simulation results offer deeper insight into the underlying physical mechanism.

Multi-thermal Dynamics in Small-Scale Active Region Phenomena: Synergies from a Coordinated 2023 SOLARNET Ground-Space Campaign

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Small-scale magnetic activity phenomena are crucial for mass and energy transport in the solar atmosphere. However, understanding their exact contribution requires analyzing the intricate coupling between localized magnetic drivers and the resulting multi-thermal plasma response. To address this, we present results from a coordinated 2023 SOLARNET observational campaign, where we captured a diverse range of fine-scale dynamics in active regions, from pervasive chromospheric and transition-region (TR) activity within the sunspot moat to a localized flaring event.

By combining high-resolution ground-based spectropolarimetry from the GREGOR telescope in the spectral window around the He I 1083.0 nm line with space-based UV observations from IRIS, we probed the multi-layer plasma response to impulsive heating. HAZEL inversions of complex He I Stokes profiles reveal significant redshifts (~30 km s^{-1}), providing clear signatures of chromospheric condensation. These dynamics are supported by simultaneous TR downflows and Mg II profile evolution observed by IRIS, confirming deep energy deposition in the lower chromosphere.

Despite clear condensation signatures, the corresponding coronal evaporative upflows often remain elusive in such compact events. Building on the legacy of observatories like Hinode, which revolutionized our understanding of localized magnetic coupling at different atmospheric levels, our 2023 campaign underscores the necessity of sustained ground–space synergies. To fully decode magnetic energy conversion, seamless temperature coverage is required. Future missions like Solar-C (EUVST) and MUSE, operating in coordination with advanced ground-based telescopes, will help to bridge the observational gap between lower-atmosphere dynamics and coronal response.

The SOLAR-C Mission in development

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1. ISAS/JAXA

The outer solar atmosphere is a highly dynamic plasma environment that produces the largest explosive events in the solar system, solar flares, which can impact human activities on Earth. The corona, much hotter than the solar surface, is heated by mechanisms not yet fully understood and is the source of the solar wind.Recent progress has been achieved through high-resolution observations, such as those by Hinode, together with numerical simulations. However, understanding fundamental plasma processes—such as waves, magnetic reconnection, and MHD instabilities—remains a key challenge.The SOLAR-C mission will address these issues using advanced spectroscopic instruments. Its unique capabilities include simultaneous observations across a wide temperature range from the chromosphere to the corona, high spatial resolution of fine structures, and sufficient cadence to capture dynamic evolution. The EUVST will enable detailed plasma diagnostics, while full-disk EUV irradiance monitoring will be conducted simultaneously.SOLAR-C is a JAXA-led international mission with contributions from the US and Europe. This presentation will provide the latest development status and ongoing efforts.

Concept of Operations for the SOLAR-C Mission

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SOLAR-C is an upcoming solar ultraviolet spectroscopic mission designed to advance our understanding of fundamental processes in solar atmospheres. Its onboard instrument, the EUV High-throughput Spectroscopic Telescope (EUVST), will perform high-sensitivity observations over a wide temperature range (10^4-10^7 K) with high spatial and temporal resolution, targeting key unresolved problems such as coronal heating, solar wind acceleration, and the energy release processes of solar flares. In addition, the Solar Spectral Irradiance Monitor (SoSpIM) will provide full-disk EUV irradiance measurements, supporting both instrument calibration and studies of solar-terrestrial interactions. To maximize scientific return, SOLAR-C adopts a flexible and responsive science operations concept, building on heritage from prior missions such as Hinode and IRIS while incorporating new technologies. Daily observation planning is led by a Chief Observer, who defines the observation timeline, including spacecraft pointing and roll angle. During the nominal operations, observation plans are generated on a daily basis to enable rapid response to dynamic solar activity. A cloud-based observation planning tool will be implemented to facilitate distributed participation and reduce the operational workload. To capture transient phenomena such as solar flares with high temporal resolution across multiple wavelengths, SOLAR-C will employ an advanced data handling scheme combining the onboard data recorder and high-speed SpaceWire communications. Data will be downlinked through a global ground station network, processed and calibrated at ISAS/JAXA and the SOLAR-C Science Center at Nagoya University, and subsequently released to the scientific community in a timely manner.

Pre-flare and active region plasma flows and structure seen by the short wavelength camera on SOLAR-C/EUVST

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The mechanisms triggering solar flares and driving coronal heating occur across wide temperature ranges on small spatial scales and short timescales, making them difficult to observe with current instrumentation. The upcoming SOLAR-C mission, launching in the late 2020s, will provide unprecedented plasma diagnostic capability with its high-throughput extreme-ultraviolet (EUV) spectroscopic telescope (EUVST), capable of ~0.2 arcsec/pix spatial sampling (~0.4 arcsec resolution), continuous temperature coverage from 0.02-15 MK, and exposure times down to 0.5 seconds. We present forward modelling of the spectrograph's short wavelength camera (170-210 Angstrom; SOLAR-C/EUVST-SW) and its response to log T~6.2 coronal plasma in a three-dimensional MHD-simulated pre-flare active region. We compare this performance to that of the previous-generation EUV Imaging Spectrometer (EIS) on Hinode (SOLAR-B). Our results demonstrate that SOLAR-C/EUVST can distinguish individual flux tubes in simulated active region loops which Hinode/EIS cannot resolve. In simulated pre-flare plasma, SOLAR-C/EUVST captures sharp velocity gradients between adjacent upflowing and downflowing plasma which Hinode/EIS is unable to resolve. Doppler velocity measurement accuracy will reach better than 1 km/s in active regions. We show that this next-generation spectrograph can be expected to directly observe processes potentially related to flare triggering, such as plasma flows from low-altitude reconnection linked to emerging flux, and determine whether active region loops consist of a small number of strands or the hundreds predicted by magnetic reconnection-induced nanoflare heating models.

Magnetic Reconnection Science Strategy for SOLAR-C

*Shinsuke Imada[1], International SOLAR-C Team[2]

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In this study, we discuss how these observational capabilities can be maximally utilized to construct a new picture of magnetic reconnection. In contrast to the conventional steady and laminar view of reconnection, SOLAR-C will enable direct observational tests of time-dependent, multi-scale reconnection involving plasmoid instability and turbulence. Specifically, it will capture the formation and disruption of fine structures together with the temporal evolution of energy release, thereby revealing the coupled processes of heating, acceleration, and transport. By integrating diagnostics of velocity, temperature, density, ionization state, and elemental composition, we aim to quantitatively constrain energy conversion efficiency, particle acceleration processes, and the role of nonequilibrium plasma in the reconnection region. In this way, SOLAR-C will bridge the gap between macroscopic reconnection dynamics and microscopic plasma processes through the combination of high spatial and temporal resolution with comprehensive plasma diagnostics. This will allow us to address key open questions, including the nature of energy conversion, the role of turbulence and plasmoid instability, and the coupling between reconnection and the surrounding solar atmosphere, ultimately establishing a new physical paradigm of magnetic reconnection in astrophysical plasmas.

Beyond Hinode: Coronal Heating Research with EUVST

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Since its launch in 2006, the Hinode mission has provided unprecedented insights into the magnetic processes that drive the dynamics, heating, and evolution of the solar atmosphere. In particular, the Extreme-ultraviolet Imaging Spectrometer (EIS) has revolutionized coronal heating studies by providing diagnostics of plasma temperatures, densities, flows, and non-thermal motions across active regions and the quiet Sun. Observations have also highlighted the need for broader temperature coverage, higher spatial resolution, greater sensitivity, and simultaneous observations of the coupled atmospheric layers. The EUV High-Throughput Spectroscopic Telescope (EUVST) on SOLAR-C is designed to address these challenges through comprehensive spectroscopic measurements spanning the chromosphere, transition region, and corona.

Connecting Coronal Loop Heating Frequency to Magnetic Properties in Non-flaring Active Regions with Hinode/EIS

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It is well-known that the heating frequency, the timescale on which energy release occurs on a given coronal strand, is variable over any given active region and that this spatial distribution of heating frequencies may vary from one active region to the next. However, a systematic study of this inter- and intra-active-region variation has not been carried out and furthermore it is not well-known how these variations are connected to the magnetic properties of the active region. In this talk, we examine the spatial distribution of heating frequencies and their correlation with the magnetic field over many different active regions and as individual active regions evolve in time. To carry out this investigation, we use raster observations from Hinode/EIS as well as imaging data from SDO/AIA to construct spatially-resolved maps of the differential emission measure (DEM) and time lag, respectively. Additionally, we carry out a series of hydrodynamic simulations constrained by field extrapolations derived from SDO/HMI magnetograms to make systematic, quantitive comparisons between the aforementioned observations and forward-modeled observables from our simulations. In doing so, we are able to map out the most-likely heating frequency across each active region raster observation. Furthermore, we explore the correlation between this distribution of heating frequencies and the changing magnetic properties of an active region as quantified by the Space weather HMI Active Region Patch (SHARP) parameters.

Effects of Plasma Composition Variations on the Radiative Cooling of Solar Flare Loops

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Plasma composition in flaring regions has been shown to have significant spatial and temporal variations, likely driven by dynamical processes that take place as a consequence of the sudden energy release at the reconnection site. The origins of these variations, as well as the effects they might, in turn, have on flare loops dynamics are not yet fully understood. In this work, we investigate the link between flare loop cooling times and plasma composition evolution in the loops formed during the M-class flare peaking at 13:56 UT on the 2022 April 2 using high cadence Hinode EIS spectroscopic observations. The analysis focuses on quantifying the cooling rate (using a series of emission lines covering a wide temperature range) and plasma composition evolution (using the Ca XIV 193.866 Å/Ar XIV 194.401 Ådiagnostic) at the apex and footpoint of the flare loop arcade. Results show slower cooling and a FIP bias of 2.4 +/- 0.2 in the loop footpoint and faster cooling and a stronger FIP bias of 2.8 +/- 0.2 in the loop apex. The potential effects of plasma composition changes on the radiative cooling process of flare loops are also investigated by comparing observed loop cooling times to those predicted by simulations from the EBTEL 0D hydrodynamic model. The EBTEL simulations show that a higher FIP bias would lead to a faster radiative cooling rate and, therefore, shorter cooling times. This suggests that the variation in FIP bias observed in the two features could be responsible for the different cooling times observed.

Power-law Relationship between Solar X-Ray Luminosity and Total Unsigned Magnetic Flux: From Active regions through X-ray bright points

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A statistical study relating coronal emission to photospheric magnetic field is one of the methods to investigate the contribution of the magnetic field to the heating of the coronal plasma. The relationship between X-ray luminosity (L_X) and total unsigned magnetic flux (Φ_m) of various heated plasma, ranging from quiet regions of the Sun through T Tauri stars, can be expressed by a single power-law, $L_X \propto \Phi_m^{1.15}$ (Pevtsov et al. 2003). In Yoshimura et al (2025), we studied the parameters dependency of the relationship for disk-integrated Sun using Hinode/XRT and SDO/HMI data, and found: [1] the relationship is characterized by a broken power-law for a wide range of the parameters, yet by a single power-law for a certain range of the cut off magnetic field strength (B_C) in the calculation of Φ_m; and [2] the power-law index is subject to variation depending on the wavelength range for X-ray integration (W_X) in the calculation of L_X.

As a next step, we examined the parameters dependency of the relationship for active regions and X-ray bright points. The relationship was found to be expressed by a single power-law with its index being approximately 1.0. The choice of B_C does not affect the power-law index; however, W_X does. The selection of shorter wavelength ranges results in higher power-law indices, which is consistent with the behavior observed in the case of disk-integrated Sun.

We will also present the preliminary results from our study on the quiet regions.

Multi-thermal analysis of loop morphology and restructuring in active regions

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Active region loops typically exhibit nearly constant cross-sectional widths along their lengths in the corona. But how this spatial structuring couples to the thermal evolution of loops that are thought to be powered by a braiding-induced nanoflare heating scenario is not fully understood. Here, we make use of a combination of high-resolution coronal observations from Solar Orbiter and SDO to address this problem. We find that coronal loops cool on timescales of only a few tens of minutes, and at the same time, their cross-sections increase. Structures that initially appear monolithic subsequently split into multiple strands, while the highest-resolution Solar Orbiter data reveal the formation of finer strands that drift apart. This behaviour is persistent across several passbands and viewing angles, suggesting that the broadening is not merely a projection effect but reflects a three-dimensional restructuring of the loops. Such morphological evolution is consistent with reconnection-driven nanoflares in which post-flare cooling leaves both thermal and structural signatures in the corona. Ongoing analysis of Hinode/EIS spectra will test whether this restructuring is accompanied by observable spectroscopic signatures (e.g., excess nonthermal width). We anticipate that future high-resolution spectroscopic observations from EUVST will provide stronger constraints on loop heating and subsequent restructuring through multi-thermal plasma diagnostics.

Coronal Rain Dynamics and Morphology under Different Coronal Heating and Magnetic Field Conditions

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1. Kyoto University

We performed 1.5-dimensional magnetohydrodynamic simulations to investigate how coronal heating and magnetic field strength regulate the dynamics and morphology of coronal rain. The occurrence of condensation critically depends on the properties of coronal heating. Many studies simplify coronal heating by assuming it to be temporally fixed and (quasi-)steady, whereas it is time-variable and exhibits both diffuse and impulsive components. The background coronal environment also plays an important role in condensation. To examine the resulting coronal response to impulsive heating, we conducted simulations along a single strand in which coronal heating is treated self-consistently and imposed a single impulsive heating event. We systematically explored two key parameters: the duration of localized heating and the strength of the coronal background magnetic fields. We find that plasma condensation becomes more likely for longer heating durations and weaker magnetic fields. The dynamics of coronal rain depend strongly on the heating duration: when the heating duration is shorter than the radiative cooling timescale, condensations preferentially fall toward both footpoints. In weak-field cases, condensation near the coronal footpoints reduces the upward energy flux from the chromosphere and suppresses coronal heating, leading to positive feedback that further promotes condensation. We also find that weaker coronal magnetic fields enhance the nonlinearity of Alfvén waves in the corona, resulting in the formation of blob-like condensations. These results highlight the importance of heating duration and coronal magnetic field strength in regulating cooling and condensation driven by impulsive heating in coronal loops.

Differences in Heating Distribution between Active-region Warm Loops and Quiet-region CBP Loops

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Based on spectroscopic data from the EIS/Hinode instrument, we investigated the heating distribution in warm loops in active regions and coronal bright point (CBP) loops in quiet regions. Using fitting formulae derived from one-dimensional numerical simulations based on a steady-state loop-heating model with the heating rate at the coronal base and heating scale length as free parameters, we estimated the ranges of the heating parameters through Bayesian inverse analysis constrained by the electron temperature and density distributions along the loops, derived from observed emission-line intensity ratios. The results indicate that heating in warm loops is concentrated near the loop footpoints, whereas CBP loops exhibit nearly uniform heating along the loops. This suggests a systematic difference in heating scale lengths between the two types of loops. We further examined brightenings observed within coronal loops identified in continuous image sequences obtained with the SDO/AIA. While brightenings in warm loops tend to occur preferentially near the loop footpoints, brightenings in CBP loops appear at arbitrary locations along the loop. Thus, the spatial distribution of brightenings appears to correlate well with the inferred heating scale lengths. For warm loops, the observed power-law relationship among heating energy flux, magnetic field strength, and loop length supports a heating model based on magnetic reconnection. Since CBP loops are also thought to be heated through magnetic reconnection, we speculate that the difference in heating scale lengths between warm loops and CBP loops may reflect intrinsic differences in the magnetic structures involved in the reconnection processes.

Modeling Elemental Abundance Variations in the Solar Corona

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Elemental abundance variations in the solar atmosphere are regularly observed in quiescent and active coronal loops, in the solar wind, and in solar flares. Elements readily ionized in the chromosphere -- elements with low first ionization potential (FIP) -- are found to be overly abundant in the corona compared to the photosphere, known as the FIP effect. Though these observations are well-known, the vast majority of modeling work has neglected these variations in both space and time. I will introduce a method to incorporate abundance variation into (magneto)hydrodynamic models, accounting for the effects on the radiative loss rate. I will show that these abundance variations have real, observable consequences, including the formation of coronal rain. I will discuss the FIP and inverse FIP effects, and how we might begin to further constrain potential causes, diagnostics, and observational implications.

Solar Soft X-ray Investigations Using CubeSat Missions

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Recent observations of solar soft X-ray emission have significantly improved our understanding of coronal heating, flare energetics, and plasma composition in the solar atmosphere. Soft X-ray measurements provide important diagnostics of multi-thermal plasma associated with active regions, solar flares, and impulsive energy release processes. The Miniature X-ray Solar Spectrometer (MinXSS) CubeSats (1, 2) series were the first solar science-focused CubeSat missions funded by the NASA Science Mission Directorate, designed to measure SXR spectral irradiance over a broad energy range (0.5–15 keV) with moderate spectral resolution (0.22 keV and 0.14 keV at 2 keV). The MinXSS missions demonstrated that CubeSat can provide scientifically valuable estimations of plasma temperatures, emission measures, and elemental abundances in the solar corona. Building on the success of MinXSS (1, 2), the Dual Aperture X-ray Solar Spectrometer (DAXSS) conducted higher spectral resolution measurements (0.09 kev) and broad energy range (0.5–15 keV) , enabling improved characterization of thermal plasma structure and elemental composition. The unique spectral observations from the CubeSats have constrained flare temperatures between approximately 6 –25 MK, quiescent coronal temperatures to 1 - 2 MK, and temporal variations of Fe, Mg, S, Si, Ar, and Ca. The increasing capability of CubeSat missions has produced scientifically valuable measurements that have complemented observations from GOES, Hinode/XRT, and SDO/AIA. Furthermore, future CubeSat missions such as the CubeSat Imaging X-ray Solar Spectrometer (CUBIXSS) and Sun Coronal Ejection Tracker (SunCET) will further advance our understanding of the solar eruptions and heating processes with new observational capabilities.

Towards a better understanding of coronal abundance anomalies: high-order multi-fluid modelling of active region loops

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Solar-C will observe the solar corona with an unprecedented large spectral coverage. However, making valuable diagnostics out of all spectral lines will require a prior understanding of some plasma properties of the observed regions, because no inversion procedure currently allows to infer all properties simultaneously in an accurate manner. Among them coronal abundances constitute a key parameter in studies of active regions, and yet they show anomalies that are poorly understood. They are usually fixed to pre-defined values that do not necessarily reflect the observed region, like when the widely used differential emission measure (DEM) is estimated. Understanding the origin of abundance anomalies has received an increased interest in the last few years, and I will start this contribution by giving a brief overview of the current theories invoked. I will then present some of the latest results obtained with the Irap Solar Atmosphere Model (ISAM), an advanced multi-fluid hydrodynamic code which has become increasingly relevant to address such problem by solving directly all species (neutrals, ions and electrons) together in a fully coupled manner. Our results show that collisions play a key role in the formation of those anomalies, where species do fractionate according to their first ionisation potential (FIP) similarly to what is seen in observations. No special tuning was necessary since the simulations are mostly parameter-free, at the exception of the loop geometry and plasma heating conditions. A question now arises on the relative importance of collision based processes versus the commonly invoked wave theory, where the goal is now to provide a unified answer with the ISAM code. Back and forth interactions between ISAM and the up-coming Solar-C observations will prove extremely useful, if not essential, to test and better quantify these processes.

Thermal and Magnetic Diagnostics of Small-scale Energy Release in Transient Brightenings of a Non-flaring Active Region Corona

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Transient brightenings (TBs) in active regions are key observational signatures of small-scale energy release in the corona. Their thermal evolution and magnetic environment provide important clues to how energy is transported and dissipated. Previous multiwavelength EUV studies have shown that TBs exhibit diverse temporal behaviors. However, how these thermal characteristics are related to the local magnetic-field structure has not yet been systematically examined. We analyzed a 17-hour period of NOAA 13738 during which no flares above C-class occurred. We identified 83 TBs using SDO/AIA and classified their temporal evolution based on the relative timing of the intensity enhancements in different EUV channels. We then compared these classifications with the local photospheric magnetic-field configuration derived from SDO/HMI, together with UV images used to identify low-atmosphere footpoint. More than half of the events showed localized mixed polarity or converging and disappearing magnetic flux near the brightening footpoint, suggesting that photospheric magnetic evolution is associated with a large fraction of the observed brightenings. However, a substantial number of events showed either no clearly identifiable footpoint or no detectable photospheric magnetic signatures despite the presence of a footpoint. The former exhibited only nearly simultaneous brightenings across multiple channels, whereas the latter was dominated by events in which high-temperature components appeared first. These results demonstrate that TBs are not thermally or magnetically uniform. The systematic correspondence between EUV temporal evolution and local magnetic environment suggests that multiple heating scenarios, possibly occurring at different atmospheric heights or spatial scales.

Active-region upflows as a source of the slow solar wind: a statistical study using Hinode/EIS and IPS observations

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The source of the slow solar wind remains one of the longstanding problems in heliophysics. Plasma upflows observed at the edges of solar active regions have been proposed as one of its candidate sources, but statistical verification over a wide range of solar activity has remained limited. In this study, we investigate whether active-region upflows contribute to the slow solar wind by combining spectroscopic observations from Hinode/EIS, interplanetary scintillation (IPS) measurements of solar-wind speed, and magnetic connectivity inferred from a PFSS model. We analyzed 51 datasets spanning more than one solar cycle from 2007 to 2022. Upflow regions were identified from EIS Doppler maps, and their magnetic connection to the source-surface solar wind was examined using PFSS field extrapolations together with IPS-derived solar-wind speed maps. Among the 51 samples, 29 showed magnetic connection between active-region upflows and open field lines reaching the source surface. Of these 29 samples, 23 were connected to slow wind with median speeds below 500 km s−1, indicating that active-region upflows can serve as a viable source of at least part of the slow solar wind under favorable magnetic conditions. Most of these connected cases were associated with open-field regions adjacent to coronal holes. We further examined the connected wind as a physical interpretation of the statistically identified source-region connection, using the Alfvén-wave-driven scaling proposed by Suzuki (2006). The upflow-connected wind is broadly consistent with this scaling only for cases with relatively small flux-tube expansion factors (ftot < 800), whereas cases with larger expansion factors deviate markedly from it. These results suggest that the acceleration of slow wind connected to active-region upflows is not uniform: Alfvén-wave-driven acceleration may be applicable to a subset of the connected wind, while additional processes may be important in highly expanded flux tubes.

The challenges of Solar Orbiter: what we have learned from an interplanetary solar physics mission so far

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Solar Orbiter has now spent 6.5 years in orbit, of which just over 4 years have been in its nominal eccentric orbit between just inside Mercury's orbit and out to that of Venus. The mission as a result has already produced over 750 refereed scientific papers tackling crucial science such as the source of disturbances in the solar wind, the smallest scales of energy yet seen in the solar corona, and the effect of solar energetic particle events on vast swathes of the heliosphere. From the point of view of conducting scientific observations, the mission has presented challenges of which a number emerged during flight. In this presentation, we revisit some of the challenges of the planning of science operations that were known before launch and some –both positive and negative –that were discovered since we started our Nominal Mission Phase.

Spectropolarimetric Inversion in Four-Dimensions (SPIn4D): Progress and Prospect

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The SPIn4D project aims to develop deep learning models that can rapidly infer 4D (space and time) MHD state variables from a temporal sequence of photospheric Stokes observations. Here I review the progress our team made so far, the lessons learned, and possible future directions. Highlights include (1) a large, publicly available library of MHD simulations and synthetic observables; (2) a physics-informed machine-learning method to resolve the 180-degree azimuthal ambiguity and calibrate the geometric height simultaneously; (3) a fast deep-learning inversion model developed for DKIST/DL-NIRSP observations.

Coronal heating by kink waves: RTV scaling laws & global models

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Since many years, it has been hypothesised that MHD waves could contribute to coronal heating, but this modelling had not gone beyond qualitative descriptions. From high-resolution simulations in the last decade of small coronal volumes we know that kink waves decay into turbulence, and subsequently heat the plasma. It has been shown numerically that this kink wave turbulent heating process can compensate for radiative losses. However, the fine resolution required for the appropriate modelling of the turbulent cascade has proven prohibitive to model the heating of large scale coronal volumes.

In this talk, I present recent modelling developments that capture the kink wave turbulent heating in a subgrid approach. This approach has allowed to construct scaling laws for the coronal loops heated with kink wave turbulence, in a similar fashion as the RTV scaling laws. These new scaling laws predict the thermodynamic properties of the kink wave heated loops, and allow for comparison with (e.g.) Hinode/EIS observations. Moreover, the subgrid approach of the turbulent heating allows to model large coronal volumes. I show our first 3D simulations of the full solar corona that is heated with kink wave turbulence.

RMHD Modeling of Reconnection Nanojets: Energy Budget, Spectral Signatures and Implications for the Solar-C Mission

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Reconnection nanojets are small-scale, short-lived jet-like bursts oriented perpendicular to the guiding magnetic field of coronal loops. They provide a direct diagnostic for small-scale magnetic reconnections in the solar corona. However, as a recently-discovered phenomenon, many fundamental questions remain unclear. Particularly, current observations have identified only a limited number of nanojet events, leaving their collective contribution to coronal heating an open question. In this talk, we will present findings on modeling nanojets based on radiative magneto-hydrodynamics (RMHD) simulations using the MURaM code. First, driven by magneto-convection, the active region corona is heated self-consistently, exhibiting fine structures such as coronal strands and coronal rain. Furthermore, as the active region evolves, a series of nanojets form - manifesting as small-scale condensations moving rapidly perpendicular to the coronal loops - which closely resemble current imaging and spectral observations. We further statistically investigate the energy partition across different temperature and kinetic energies of these nanojets. By producing synthetic spectra for the upcoming Solar-C mission, we forward model the observational signatures of these events and demonstrate their decisive role in coronal heating. Finally, we will introduce which spectral lines and viewing angles are most effective for identifying these nanojets in future Solar-C/EUVST observations.

Spectral Line Ratios: A Diagnostic of Elemental Abundances

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We observe that elemental abundances in the corona change over space and time, known as the First Ionization Potential (FIP) effect. In ultra violet wavelengths, these abundances are inferred through spectral line ratios. Line ratios are very sensitive to temperature, so certain line ratios may be more reliable than others. Using a hydrodynamic model that allows elemental abundances to change with space and time, we can compare simulated abundance factors to ones derived observationally through line ratios. We examine the spectral line ratios observable by instruments such as Hinode/EIS and EUVST, and we suggest which line ratios might be most reliable.

Faint Hot Components of Quiescent Active Regions: a Statistical Study with NuSTAR, XRT, and AIA

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Active regions contain plasma across a broad range of temperatures, with the peak of the thermal distribution typically in the few MK. Prior studies have identified evidence for a faint, much hotter component in a few quiescent regions (>5 MK, even >10 MK) (e.g. Ishikawa et al. 2017). Such components are a powerful diagnostic of the frequency of heat input leading to the elevated temperature of the corona, but they are challenging to observe. Only direct-focusing, spectrally-resolved x-ray instruments are sufficiently sensitive to hot sources as faint as those from quiescent regions. Sounding-rocket-based measurements from *FOXSI* and *MaGIXS* have allowed characterization of hot components for a few regions, but are fundamentally limited in observation time. The *NuSTAR* astrophysical x-ray telescope possesses sufficient sensitivity to characterize these components, and is less time-limited. It has conducted more than 100 hours of solar observation, generating a catalog of more than 30 active regions from 2014-2023. Building on prior work that examined one of these, we present a statistical analysis of the thermal distribution of all *NuSTAR* active regions. We conduct DEM analysis using *NuSTAR*, XRT (when available), and AIA, and discuss the prevalence of faint, hot components. Finally, we discuss the implications of these results with respect to coronal heating.

Compression, Impact and Hot Rebound Flows from Coronal Rain Downflows

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1. Northumbria University

Studying coronal rain formation through thermal non-equilibrium (TNE) and thermal instability (TI) provides insights into coronal heating mechanisms. We analysed a quiescent coronal rain event using space-based observations from the High-Resolution Imager in Extreme Ultraviolet (HRIEUV) of Solar Orbiter (SolO), the Atmospheric Imaging Assembly (AIA) of the Solar Dynamics Observatory (SDO), and the Slit-Jaw Imager (SJI) from the Interface Region Imaging Spectrograph (IRIS) from November 1st, 2023. During the coronal rain shower, the coronal loop exhibits substantial EUV variability and structural changes. Rain clumps fell at 72–87 km s^{-1} with cool EUV absorbing core sizes of ≈600 km and densities of ≈6 $\times 10^{11}$ cm^{-3} preceded by strong compressions. These mostly isothermal compressions suggest energy transfer into the rain, decelerating it and possibly reducing cooling rates −consistent with accretion braking timescales. The shower carried microflare-level energy (4.64 $\times 10^{26}$ erg), with clumps producing impacts that reach the lower transition region and are visible across all EUV channels and in SJI 1400 Å. The impacts generated hot rebound flows ($10^{6.2}$–$10^{6.3}$ K, 85–87 km s^{-1}) that refilled and reheated the loop but carried less than 15% of the clumps’ kinetic energy. We detected steady footpoint heating signatures consistent with the TNE-TI scenario, with an estimated amplitude of $10^{-2\pm0.3}$ erg cm^{-3} s^{-1} and heating scale heights of 2–10 Mm, matching active region values. Coronal rain may thus serve as both a template for accretion braking and a proxy for integrated heating driving TNE-TI cycles.

A Bayesian Approach to Assessing the Accuracy of Coronal Abundance Diagnostics

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Elemental composition is a key diagnostic for understanding the physical processes that shape the solar corona. Variations in elemental abundances across solar structures arise from the First Ionization Potential effect, making accurate abundance measurements essential for studying coronal heating, plasma transport, and magnetic structuring. In this work, we evaluate and compare several established techniques for determining relative elemental abundances using EUV spectroscopy from the Hinode Extreme Ultraviolet Imaging Spectrometer, EIS.
We apply four diagnostic approaches, including line ratios, differential emission measure inversions, and the Linear Combination Ratio technique, to a common set of EIS spectral lines. By analyzing their performance side by side, we assess the consistency, sensitivity, and practical limitations of each method when applied to real solar observations.
To quantify the reliability of the inferred abundances, we introduce a Bayesian framework that evaluates both measurement accuracy and robustness to observational noise. This probabilistic approach provides a structured way to estimate uncertainties and to identify conditions under which each diagnostic performs optimally.
All methodologies developed in this study are designed to be fully extendable to EUVST on board Solar C.

Hinode/EUV Imaging Spectrometer Observations of the Sun-as-a-Star

*David H Brooks[1]

1. Computational Physics, Inc.

Hinode/EIS regularly performs a program of full disk mosaic observations in coordination with IRIS (HOP 344). The EIS data have been used, for example, to construct a map of the sources of the slow solar wind, as a resource for statistical samples of active regions, and for testing future observing schemes to cross-calibrate Solar-C/EUVST and SoSPIM. Here we report on a novel application of this observing program to study the performance of EIS as a Sun-as-a-star EUV spectrometer. Previous full Sun observations from SDO/EVE have shown that coronal composition is highly correlated with the F10.7 cm radio flux, and therefore varies with the solar cycle. We construct full disk averaged EIS spectra for the period 2013-2022, and use the Si X 258/S X 264 elemental abundance diagnostic to confirm this trend with a high correlation coefficient (r=0.91). We also use the Fe X 257A magnetically induced transition (MIT) to measure the strength of the coronal magnetic field in the Sun-as-a-star, finding a weak field value of ~2G (below the sensitivity of the diagnostic).

Deciphering filament eruptions and flare energy release through coordinated observations with Solar Orbiter

*Miho Janvier[1]

1. European Space Agency

Filament eruptions are among the most dynamic manifestations of magnetic energy storage and sudden release in the solar atmosphere. Their destabilization can trigger solar flares and is a source of coronal mass ejections, making them key phenomena for understanding how magnetic energy is dissipated during these intense events and how the Sun’s variability impacts objects in the solar system.

Observations from missions such as Hinode and ground-based observatories have revealed the importance of multi-thermal plasma evolution, current-sheet formation, and magnetic reconnection across the different layers of the Sun’s atmosphere. Yet major questions remain regarding the onset mechanisms of eruptions, the coupling between different atmospheric layers, and the extent to which eruptive structures maintain their magnetic and compositional signatures as they propagate into the heliosphere.

Recent coordinated campaigns involving Solar Orbiter, together with near-Earth observatories and ground-based facilities now offer an unprecedented opportunity to investigate eruptive events from multiple viewing angles and across a broad range of spatial and temporal scales. In this presentation, I will discuss recent case studies of filament eruptions and associated flare activity observed with Solar Orbiter and other assets. These results demonstrate how coordinated observations provide new perspectives on the observational strategies and diagnostics that will be essential for the next generation of solar missions, including Solar-C.

Fine structure of flares as revealed by Solar Orbiter/EUI.

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Since 2023, the Solar Orbiter's high-resolution instrument suite has executed a flare-tailored operational mode during remote sensing windows at orbital perihelia. By interleaving sequences of short-exposure frames taken at 2-second cadence with normal-exposures, the Extreme Ultraviolet Imager's High Resolution Telescope (HRI_EUV) captures bright, typically saturated flare ribbons while maintaining active region context. This mode reveals unprecedented sub-structures that advance our understanding of flare energy release and particle acceleration. This presentation details the spatial and temporal scales of these sub-structures (EUV kernels) within an M2.5 GOES-class flare observed at 0.38 AU. The flare kernels are remarkably small (< 1 Mm^2)—with a significant fraction remaining unresolved despite Solar Orbiter's proximity to the Sun—and exhibit surprisingly short heating timescales of less than a few seconds. We explore the implications of these localised, transient energy injections on the total flare energy flux and the resulting atmospheric response. Finally, we introduce a comprehensive, publicly available catalogue of ~50 flares captured during these high-cadence campaigns, including coordinated observations with Hinode, and we highlight initial insights gained by leveraging the entire dataset. These takeaways from Solar Orbiter provide a vital framework for next-generation flare-observing satellites, directly optimising the science return for the upcoming Solar-C mission.

Flare Fine Structure: Recent 3D Modeling Results and Opportunities for Solar-C

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Solar flares are spectacular manifestations of explosive energy release powered by magnetic reconnection. While the standard CSHKP model has proven highly successful in explaining key features of flare observations in terms of magnetic reconnection, many aspects of the energy release are not yet understood. Although direct diagnosis of the magnetic field dynamics in the corona remains highly challenging, rich information may be gleaned from flare ribbons, which represent the chromospheric imprints of reconnection in the corona. We present results from recent high-resolution 3D flare modeling that reveals how turbulent, bursty reconnection dynamics may imprint fine structure on flare ribbons. In particular, we show that current sheet plasmoids manifest as transient ‘spirals’ along the flare ribbon front and show how such features may be used to constrain quantitative properties of the current sheet dynamics. We discuss the implications of these results for present and future high-resolution solar imaging observations.

Spatial and temporal variations of energy transport during solar flares: looking forward to Solar-C

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Solar flares exhibit fine scale structures, spatially, temporally, and spectrally. It is very likely that this sub-structure carries important information about variations of energy transport properties. However, extracting that information is not always straightforward. I will present a few examples of solar flare model-data comparisons that present interesting challenges our understanding of flare processes. That includes results from Hinode/EIS, Hinode/SOT, IRIS, and from Solar Orbiter's Major Flare campaign observations. The latter in particular reveals exciting diagnostic potential of Solar-C/EUVST observations during the next solar cycle.

Solar flare footpoint line broadening is predominantly field-aligned

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Magnetic reconnection powers solar and stellar flares, but a full understanding of how the released energy is transported and converted within the solar atmosphere remains elusive. One clue lies at solar-flare footpoints, where spectral lines are far broader than the electron temperature alone can explain. Unresolved flows, waves, turbulence and ion heating have all been proposed, but observations have not yet conclusively distinguished between these mechanisms. Here we perform an unprecedented geometric test for flare footpoints, using 4,593 Hinode/EIS spectra from 407 C- to M-class flares. Line widths decrease systematically from disk centre to limb in all coronal emission lines, showing that the dominant broadening component is magnetic field aligned rather than isotropic or transverse. Cooler lines retain substantial broadening into the early decay phase, consistent with persistent unresolved field-aligned flows or line-of-sight velocity gradients. Hotter lines show an impulsive component that decays rapidly after the soft X-ray peak, consistent with preferential parallel ion heating. Together, these results answer a central question in flare spectroscopy. The excess flare-footpoint line width is dominated by field-aligned contributions from persistent unresolved flows or velocity gradients in cooler plasma and an impulsive component consistent with parallel ion heating in the hottest emission.

Analyzing the hottest plasma in solar flares using X-ray measurements

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Using X-ray observations provides access to the hottest plasma components in solar flares. In particular, hard X-ray (HXR) measurements of large (X-class) flares reveal an additional high-temperature thermal component, commonly referred to as the *superhot* component, with temperatures typically above 30 MK (e.g., Caspi et al. 2010). Its origin remains uncertain, although it has been proposed to be a signature of direct heating at the magnetic reconnection site in the solar corona.

The Spectrometer/Telescope for Imaging X-rays (STIX) onboard Solar Orbiter has observed over 100 X-class flares, offering a unique opportunity to investigate the hottest plasma evolution through combined spatial and spectral analysis within a multi-thermal framework. In this work, we present recent observations of X-class flares seen by STIX in combination with other X-ray including XRT/Hinode.

We find that the superhot component is present in most of the analyzed events with the superhot component containing about 20% of the total flare energy at flare peak time. Applying the spectral component imaging technique (Stiefel et al. 2025), we identify a spatial separation and delay between plasmas at different temperatures supporting the idea that superhot sources are linked to direct heating located above the flare loops.

A long-term statistical analysis of cool coronal material linked to magnetic complexity and flaring

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The study of coronal rain and other cool materials observed in the Sun’ s upper atmosphere is becoming increasingly important given their connection to the fundamental mechanisms of coronal heating, flaring and eruption. The widely accepted formation mechanism of the phenomena is through concentrated footpoint heating of coronal loops. This heating leads to a cyclical process known as the Thermal Non-Equilibrium - Thermal Instability (TNE-TI) cycle, resulting in an over-density of material, causing a subsequent rapid cooling of the plasma and its condensation into coronal rain. The drivers of this footpoint heating are still unknown though it is generally hypothesised that there may be a strong link to the local magnetic complexity.

We present a large-scale study of the solar atmosphere spanning three years of SDO observations to determine various statistics related to the amount and distribution of cool coronal material observed above active regions on the Sun. In particular, we look into the relation between the amount and height distribution of cool coronal material and the magnetic complexity of an active region. We also look for any relation between the flare-driven coronal rain and the properties of the flare at its origin. These statistics are achieved utilising the AIA 304 channel processed with the DeepFilter algorithm. We briefly present our new machine learning based algorithm, designed to morphologically detect the cool material emission within the 304 passband.

Modeling Flare Continuum Emission Observed by Hinode/EIS: Instrument Calibration and Element Composition Results

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Continuum emission from a solar flare observed with the Extreme ultraviolet Imaging Spectrometer (EIS) on board the Hinode satellite is used to obtain the radiometric calibration of the instrument. The flare had a GOES class of M8, and peaked at 23:59 UT on 2024 September 30. The continuum is modeled by computing a differential emission measure curve using EIS emission lines and atomic data from the CHIANTI database. The ratio of the observed continuum to model continuum yields effective area curves for the instrument. The new curves confirm earlier findings that the EIS long-wavelength channel has degraded by a factor two compared to the short-wavelength channel. However, no evidence is found for the fine-scale structure in the effective area curves that has been presented by previous authors. In order to reproduce both the emission line intensities and the continuum, it is found that the plasma must be depleted in elements with low first ionization potentials (FIPs), i.e., the so-called inverse FIP-effect. In particular, the Fe/H relative abundance is found to be a factor 0.57 below the photospheric value at a temperature of 10 MK. This is confirmed by analysis of soft X-ray spectra from the Solar X-ray Monitor on Chandrayaan-2, which yields an Fe/H FIP bias of 0.55 averaged over the entire flare.

Investigating Quasi-Periodic Pulsations in Solar Flares with Solar Orbiter/EUI Short-Exposure Images

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Quasi-periodic pulsations (QPPs) are nearly periodic oscillations frequently observed in irradiance time series during solar flares, and have also been reported in several stellar flares. The most likely mechanisms proposed to explain these fluctuations include spontaneous or externally driven periodic reconnection, as well as MHD waves. Disentangling these possible origins, however, remains challenging due to the lack of appropriate observations.

Among the reported periods, oscillations in the 10–30 s range are regularly reported in full-Sun integrated time series, largely independently of flare size or amplitude, and have often been associated with periodic reconnection processes. Confirming the origin of these fluctuations nevertheless requires high-cadence, spatially resolved observations that remain unsaturated in the flare core regions, datasets that are rarely available.

The EUI instrument onboard the Solar Orbiter spacecraft regularly implements a dedicated observing strategy alternating standard observations with short-exposure images acquired at cadences of 2 s. While the nominal-exposure images reveal the full flare environment, the short-exposure observations isolate the brightest emission sources without saturation. Combining both types of observations allows flare ribbon kernels as well as fine flare-loop structures to be resolved, whose spatial distribution and temporal evolution provide valuable new insight into the mechanisms underlying QPPs, as demonstrated in this presentation through several case studies.

These observing campaigns also provide a strong basis for preparing the scientific exploitation of Solar-C, particularly for the SOSPIM instrument, for which QPP studies will represent a natural science objective.

Comparing Observed and Synthetic Spectropolarimetric Signatures of Solar Flares with Machine Learning

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Spectropolarimetric observations of solar flares provide key insights into the thermodynamic and magnetic properties of the lower solar atmosphere. To understand the broader picture of these events, researchers typically employ a multiwavelength approach, tracing processes from the photosphere up to the corona. The interpretation of these complex observations can rely on advanced 3D radiative magnetohydrodynamic (MHD) simulations. However, validating these models across the wide range of atmospheric layers they encompass remains a significant challenge. While validations of simulated flaring events frequently focus on the corona and transition region, extending forward modeling to the lower atmosphere offers a more comprehensive view of the physical coherence throughout the entire domain. For this type of model, moving toward the validation of direct physical observables, such as full Stokes profiles from the photosphere and chromosphere, provides a unique and valuable perspective. In this contribution, we present a data-driven framework based on K-means clustering to establish a connection between large-scale spectropolarimetric observations and state-of-the-art numerical simulations of the lower atmosphere. By applying this unsupervised machine learning technique to full Stokes vector profiles, both observed and forward-modeled synthetic spectra are projected onto a common standardized domain allowing for a direct and self-consistent comparison between observations and simulations. Through this unified clustering analysis, we identify common atmospheric signatures and assess how accurately simulations reproduce energy release events. Overall, the joint clustering analysis of observed and synthetic spectropolarimetric datasets provides a powerful framework for testing theoretical models and investigating the underlying physics of complex solar phenomena.

Direct Focusing X-ray Imaging Spectroscopy of Solar Flares with FOXSI-4 and FOXSI-5

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The Focusing Optics X-ray Solar Imager (FOXSI) is a US-Japan collaborative Solar-dedicated sounding rocket experiment that aims to realize direct focusing imaging spectroscopic observation in soft and hard X-ray up to 20 keV. FOXSI has been successfully launched three times, targeting the quiet Sun, active regions, and microflares. In contrast, the fourth flight, FOXSI-4, was conducted as part of NASA's first Flare Campaign, in which the payload was kept on standby for rapid launch during a two-week observing window to target a mid-to-large class solar flare (>GOES C5 class). FOXSI-4 incorporated several major hardware upgrades, including high-resolution Wolter-I optics, CMOS soft X-ray detectors, CdTe hard X-ray double-sided strip detectors, and a Quad-Timepix3 detector. FOXSI-4 was successfully launched from Poker Flat Research Range, Alaska, on April 17, 2024. During the flight, FOXSI-4 successfully observed an active region (for about 1 minute) and, for the first time, a GOES M1-class flare (for about 3.5 minutes, decay phase). These results demonstrate the technical readiness of direct-focusing optics for future space-based observations of soft and hard X-ray emission from solar flares.To enable quantitative analysis of the FOXSI-4 flare observation, we are developing an analysis framework that integrates event reconstruction, pointing information, data selection, and spectral fitting. We have also constructed response functions for various combinations of attenuators, optics, and detectors, and implemented a GUI-based data reduction tool and an XSPEC-based spectral fitting pipeline for quantitative spatially and temporally resolved spectroscopy. Following this success, FOXSI-5, a re-flight mission based on the FOXSI-4 configuration with a few upgrades and replaced components, has been selected and is scheduled for launch from Alaska in May 2026. This presentation will report the status of the FOXSI-4 flare data analysis, as well as the launch preparation, ground testing, and initial observational results of FOXSI-5.

Characterizing Flare Eruptivity of Solar Active Regions Based on Photospheric Magnetic Properties and Coronal Modeling

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Discriminating between eruptive and confined solar flares based on photospheric magnetic data of active regions remains a significant challenge in solar physics. This is because single-parameter models often encounter limitations in capturing the actual physical thresholds of eruptions. In this study, we further evaluate the eruption onset of solar flares, building upon the framework developed by Muhamad & Kusano (2025) by evaluating a nondimensional metric that combines electric current neutralization with torus instability (TI) analysis. Using SDO/HMI vector magnetograms and nonlinear force-free field modeling for large solar flare events, we examine the methodological challenges of estimating the critical height within the complex environment of active regions. We compare decay index profiles derived from maximum free-energy proxies against estimations along the polarity inversion line, while considering how observational factors affect the determination of eruption thresholds. Our analysis suggests that while noise in photospheric boundary data can influence the identification of critical heights, the values derived from coronal modeling are generally consistent with theoretical expectations and stereoscopic observations. These results indicate that accounting for both current neutralization and coronal modeling provides a more comprehensive basis for characterizing the eruptivity of solar flares.

Solar Flare Ion Temperatures

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This talk argues that the ion temperature is several times the local electron temperature in the hot onset phase and in the above-the-loop region of solar flares. The talk considers: (1) spectral line Doppler widths (“nonthermal” broadening) measured by Hinode, IRIS, Yohkoh and earlier solar missions; (2) evidence for “universal” ion and electron temperature increase scaling relations for magnetic reconnection in the solar wind, Earth’s magnetopause, Earth’s magnetotail, and numerical simulations; and (3) thermal equilibration times for onset and above-the-loop densities, which are much longer than previous estimates based on soft X-ray flare loops. We conclude that the ion temperature is likely to reach 60 MK or greater and that it may represent a substantial part of spectral line widths, significantly contributing to solving the long-standing issue of the excess line widths in hot flare lines.

Related publication: Russell et al. 2025, ApJ Letters, 990:L39.